

Short-term load forecasting based on LSTM networks considering attention mechanism

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ABSTRACT

Reliable and accurate zonal electricity load forecasting is essential for power system operation and planning. Probabilistic load forecasts can present more comprehensive information for decision-making processes by quantifying the uncertainties of the electric load. A suitable feature selection is a critical step in forecasting, especially for data-driven methods. Weather conditions are another major factor related to electricity demand and play an important role in load forecasting. In this paper, we propose a dual-stage attention based long short-term memory (LSTM) network for short-term zonal load probabilistic forecasting. In the first stage, a feature attention based encoder is built to calculate the correlation of input features with electricity load at each time step. The most relevant input features can be adaptively selected. In the second stage, a temporal attention based decoder is developed to mine the time dependencies. Then, an LSTM model integrates these attention results and the probabilistic forecasts can be obtained using a pinball loss function. We also discuss how the proposed method can be utilized for feature and weather station selection. The effectiveness of the proposed method for both point and probabilistic forecasting is adequately verified on an open dataset of GEFCom2014, showing higher accuracy and generalization ability over other state-of-the-art forecasting models.

1. Introduction

Accurate load forecasting has always been crucially important during the modern power system developments. It provides essential information for the economic operation, scheduling electricity transactions and assessing the system security. Particularly, the forecasting horizon for short-term load forecasting is from several minutes up to two weeks [1]. A reliable short-term load forecasts is helpful for utilities and system operators to make decisions. However, there exists significant uncertainties of the load profiles arising from the climate variability and distributed renewable energy sources (DRES) [2]. The stochastic nature of DRES, such as the photovoltaics, makes the load harder to forecast [3]. Unfortunately, the DRES are always installed behind the meters, and their power generations are always invisible to utilities and system operators. Therefore, it is challenging to build a model to accurately predict the zonal load and capture the uncertainties only with the historical zonal load measurements and weather condition data.

Most of the load forecasting approaches proposed by researchers focus on point forecasting, which can be divided into three categories:

statistical models, artificial intelligence models and hybrid models, such as autoregressive integrated moving average (ARIMA) model [4], semi-parametric additive model [5], support vector regression model [6], decision tree-based model [7] and deep neural networks [8–9], etc. Comprehensive reviews and evaluations of existing methods can be found in [10,11]. However, in consideration of capturing the uncertainty of future load fluctuations, point forecasting is not enough because it only provides a single value at each time step for decision making. An ideal forecasting model should present more information in the form of quantiles, intervals or density functions in the dynamic environment. For that reason, probabilistic load forecasting has received extensive attention.

Probabilistic load forecasting models can be built via three components according to [1]: generating multiple scenarios as the inputs of a point forecasting model [12–14–15], utilizing a probabilistic forecasting model [16–17–18], and post-processing the point forecasts result [19]. Hyndman and Fan [12] proposed a long-term peak load density forecasting method considering the effect of a range of uncertainties of the external environment. They first used a semi-parametric additive model to obtain the relationships between the load and the driver

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variables, such as temperatures and calendar effects, etc. Then the demand distributions can be forecasted through simulation of temperature and assumed future economic scenarios. However, they did not give a specific degree of correlation between those variables. Xie *et al.* analyzed the influence of different weather variables with generated temperature and humidity scenarios on the performance of forecasting model [13,14]. Hong *et al.* presented a probabilistic forecast model through cross scenario analysis with hourly information [15]. Wang *et al.* integrated long short-term memory network (LSTM) with a pinball loss function to forecast individual load [16]. Chen *et al.* presented a modified deep residual network to improve the forecast results. They also used an ensemble strategy to improve the generalization capability of their model [17]. Zhang *et al.* proposed an improved quantile regression neural network for probabilistic load forecasting. They adopted a series of model optimization methods including dropout, early stopping and batch training to prevent the over-fitting problem [18]. Liu *et al.* generated probabilistic load forecasts by quantile regression averaging on a set of sister point forecasts, without relying on high-quality expert forecasts [19]. A comprehensive review of probabilistic load forecasting is presented in [1].

The accuracy of probabilistic forecasting is affected by some exogenous factors, such as weather, holidays and economic information, etc. The conventional methods focus one or more of these variables as inputs of the models on the basis of expert knowledge without appropriate variable selection. Although more input variables could contain more reference information in general, excessive variables may bring more shortcomings in practice, especially for the neural networks which is greatly dependent on the scale of inputs. As a result, these random assignments for the model inputs may bring irrelevant parameters, leading to unreliable performance. Moreover, the over fitting and time-consuming issues may arise if there are a fair number of redundant inputs. Variable selection is the process to select optimal input features with the highest correlation to the outputs. Recently, several data-driven approaches have been developed for variable selection to reduce the complexity of forecast models including dimensionality reduction methods and output correlation analysis methods. Dimensionality reduction methods select the dominant variables by removing the variables with high similarity. [20] ranked the quality of attributes on the basis of their instance values via calculating the distances between them, and then randomly selected K-nearest neighbors from the same class. In [21], an orthogonal forward regression was used to rank the candidate variables in the order of decreasing relevance after randomly shuffling the components of these vectors. Another index of ranking based on conditional mutual information was proposed in [22] to estimate the quantity of dependence of two variables. Despite these dimensionality reduction methods can lower the size of inputs and computational cost, they cannot guarantee a better forecast result because there may exist some connections between the reduced variables and forecast outputs in a certain scenario. For example, temperatures play a key role for load forecasting in summer and winter, while humidity and wind speed is dominant in late Spring and early Autumn [13]. By contrast, the output correlation analysis methods, which are decided by the target problem and consider the influence of input variables on the outputs, can better select the key variables and improve the accuracy at the same time. A holistic and heuristic variable selection method based on probabilistic error measures and point errors to reduce the generated temperature scenarios is proposed in [23]. [24] adopted the least absolute shrinkage and selection operator (LASSO) to automatically select variables, in which the coefficients of parameters were varying over time. A method combined LASSO and penalized regression was applied in [25]. The output correlation analysis methods can help yield accurate prediction results to some extent. However, the entire process of variable selection is still a black box, which is lack of explanatory ability. In actual applications, researchers can set some constraints and get the right variables through a lot of experiments, but the reasons behind for variable selection cannot be adequately explained. Therefore, there is a critical

need to propose an intelligent variable selection method to distinctly show the exact impact of different variables on the objective load forecasting. In recent years, with the development of machine learning techniques, a series of deep learning models have been proposed, such as backpropagation neural network (BPNN), convolutional neural network (CNN) and long-short term neural network (LSTM). The deep learning models have already shown their success for solving the classification and regression problems which are attribute to their superior ability of extracting non-linear hidden features. Among these deep learning models, the LSTM networks are more suitable for dealing with time series problems. However, LSTM models suffer a main drawback that they intrinsically capture the temporal variations of the input time sequential data, whereas the correlations between the different input features are ignored. To this end, an innovative architecture, namely encoder-decoder, is proposed for better modeling the hidden features, considering both the related and temporal characteristics of inputs [26].

Moreover, few studies have focused on improving zonal load forecasting by means of suitable variable selections. Usually, there are multiple weather stations in a region to hourly collect data. As mentioned in [11] and [1], it is of vital importance to study how to utilize these weather data to improve the load forecast accuracy. To the best of our knowledge, [27] was the first attempt to solve this problem using a simple framework. After that, it appears that no work has been done to give a better solution for figuring out which and how many weather stations to use for zonal load forecasting.

Motivated by uncertainties existing in demand load and multiple data sources obtained for zonal forecasting, an end-to-end framework of dual-stage attention based quantile long short-term memory network (DA-QLSTM) is presented to forecast short-term electricity load in this paper. Our work improves the method of [28] in two aspects. The first is we adopt the pinball loss as the loss function to obtain the probabilistic forecasting results, which can better capture the load uncertainties and present more valuable information for distribution system planning and operation. The second is a quantified measure based on the attention scores is presented to select the relevant weather station to further improve the zonal load forecasting accuracy. In the first stage of our proposed model, we develop a feature attention based encoder to calculate the correlation of input features with the electricity load at each time step. The most relevant input features can be adaptively selected. In the second stage, a temporal attention based decoder is built to capture the temporal dependencies, and select relevant encoder hidden states through all time steps. Then we use an LSTM model to integrate these final attention results and jointly train the model. Compared with static variable selection in the past work, both of the time varying and sequential correlation with object forecasts are taken into consideration to better characterize the dynamic influence of input variables. In order to obtain the probabilistic forecast results, a novel pinball loss function is used instead to present more uncertainty information.

The main contributions of this work are summarized as follows:

- 1) A dual-stage attention based LSTM network is proposed for short-term zonal load forecasting. Both feature correlation and temporal dependencies are considered and quantified, which are instructive for variable selection
- 2) By adaptively enhance the impact of higher correlation features and reduce the impacts from the irrelevant variables, the proposed model has superior robustness under different weather conditions.
- 3) For zonal load forecasting, the problems of selecting which and how many weather station data for the model inputs are well solved. The forecasting accuracy at the zonal level are further improved by removing the constraint on the fixed number of weather stations.
- 4) Comprehensive comparisons with various state-of-the-art methods are conducted, which demonstrate a superior performance of the proposed method with respect to both the point and probabilistic forecasting results. Even without some complex feature engineering, the proposed methods still remain accurate and robust, which can relieve

the burden on system operators to some extent.

The rest of the paper is organized as follows: Section II introduces the framework of a dual-stage attention based LSTM. Section III presents the framework of short-term load probabilistic forecasting. Evaluation metrics and benchmarks are described in Section IV. Section V presents the comprehensive case studies on the GEFCom2014 datasets. Finally, conclusions are drawn and an outlook of future implications and extensions are discussed in Section VI.

2. Dual-Stage attention based LSTM

In this section, the feature attention based encoder and temporal attention based decoder are demonstrated on how they enhance and suppress the information among the input features and the time points.

2.1. Feature attention based encoder

As mentioned above, the main task of the encoder is to quantify the impacts of the input features on the target series. Using the extracted correlation on the features, we could replace the original input with the most relevant features for hidden units reduction.

Given historical data of a target series $Y = (y_1, y_2, \dots, y_T) \in \mathbb{R}^T$ and exogenous feature time series $X = (x_1, x_2, \dots, x_T) \in \mathbb{R}^{M \times T}$, we use a feature attention based encoder to construct a weighted feature representation to replace the original input. The structure of the feature attention based encoder is shown in Fig. 1.

The feature matrix is composed by the measured M features at T time points. We first split the matrix into M vectors, where $x_\tau = (x_\tau^1, x_\tau^2, \dots, x_\tau^M)$ represents a vector of M feature items at time point τ , and $x^m = (x_1^m, x_2^m, \dots, x_T^m)$ presents the m -th feature series time of length T in the history.

In essence, the encoder is built on a recurrent neural network (RNN), so as to introduce the feature impacts into the whole historical series. The traditional RNN is trained by backpropagation through time on hidden states that capture temporal dependencies. However, its performance deteriorates rapidly since the large time scale often leads to gradients vanishing or exploding during the model training. To address this issue, the long short-term memory network (LSTM) is developed on RNN such that a memory cell and three controlled gates are additionally included within each LSTM basic recurrent unit. Therefore, the LSTM network is selected to capture time dependencies under the feature series and wrap the attention weight during time iterations.

For each iteration in LSTM, the current input x_τ is concatenated with the previous hidden state $h_{\tau-1}$, and then passes through three fully

connected gates in the LSTM unit. For the forget gate f_τ^e , only a part of the information from the last cell state $s_{\tau-1}^e$ is retained, while the rest are abandoned. The input gate i_τ^e selects the aspects of current input information to be preserved. The cell state is updated to s_τ^e by the summation of information transferred through forget gate and input gate. Then the output gate o_τ^e decides which information in the updated cell state s_τ^e could pass on as the final output for the next time step. Briefly, the process of the encoder LSTM network is formulated as follows:

$$f_\tau^e = \sigma(W_f^e[h_{\tau-1}; x_\tau] + b_f^e) \quad (1)$$

$$i_\tau^e = \sigma(W_i^e[h_{\tau-1}; x_\tau] + b_i^e) \quad (2)$$

$$o_\tau^e = \sigma(W_o^e[h_{\tau-1}; x_\tau] + b_o^e) \quad (3)$$

$$s_\tau^e = f_\tau^e \odot s_{\tau-1}^e + i_\tau^e \odot \tanh(W_s^e[h_{\tau-1}; x_\tau] + b_s^e) \quad (4)$$

$$h_\tau = o_\tau^e \odot \tanh(s_\tau^e) \quad (5)$$

where W_f^e , W_i^e , W_o^e and b_f^e , b_i^e , b_o^e are the gate weight matrices and biases of the forget, input and output gates, respectively; W_s^e and b_s^e are the weight matrix and bias of the memory cell; $\odot$ represents the element-wise multiplication; σ is the sigmoid activation function and $[\cdot]$ represents the matrix concatenation.

Firstly, every input feature series is put through a deterministic attention model so as to quantify how close is the relationship between each feature and the target attribute. A two-layer perceptron is constructed for the attention score calculation in Eq.6. In addition to the feature series itself, historical information under the previous hidden state $h_{\tau-1}^e$ and the cell state $s_{\tau-1}^e$ of the encoder LSTM network are also considered during the attention score calculation.

$$e_\tau^m = V_e^T \tanh(W_e[h_{\tau-1}; s_{\tau-1}] + U_e x^m + b_e) \quad (6)$$

where $V_e \in \mathbb{R}^T$, $W_e \in \mathbb{R}^{T \times 2p}$, $U_e \in \mathbb{R}^{T \times T}$ and $b_e \in \mathbb{R}^T$ are weights and bias of the two-layer perceptron in encoder, and p is the number of hidden units in the last layer of encoder LSTM network.

The attention score e_τ^m is normalized by the softmax function as the following:

$$\alpha_\tau^m = \text{softmax}(e_\tau^m) = \frac{\exp(e_\tau^m)}{\sum_{i=1}^M \exp(e_\tau^i)} \quad (7)$$

Thus, the feature attention weight α_τ^m , whose summation on m is 1, is positively related to the significance of the m -th feature at the time step τ .

As mentioned previously, we attempt to form a new feature representation that focuses on the most relevant features. This new expression form $\tilde{x}_\tau$ is defined by the multiplication of α_τ^m and the original feature input x_τ^m as shown below:

$$\tilde{x}_\tau = (\alpha_\tau^1 x_\tau^1, \alpha_\tau^2 x_\tau^2, \dots, \alpha_\tau^M x_\tau^M)^T \in \mathbb{R}^{1 \times M} \quad (8)$$

The implication is that whether the information of the m -th feature at the time step τ is to be enhanced or to be suppressed is determined by the value of the attention weight α_τ^m .

Therefore, the new weighted feature representation $\tilde{x}_\tau$, rather than its original form x_τ , is fed into the encoder LSTM network for the iteration.

The output of the encoder is the set of the hidden states of the encoder LSTM network at each time step as $\{h_\tau\}_{\tau=1}^T$, and each hidden state h_τ denotes the selective expression of features at the time step τ as the following:

$$h_\tau = f_{LSTM}^e(h_{\tau-1}, \tilde{x}_\tau) \in \mathbb{R}^{1 \times p} \quad (9)$$

where f_{LSTM}^e represents the function set of the encoder LSTM network

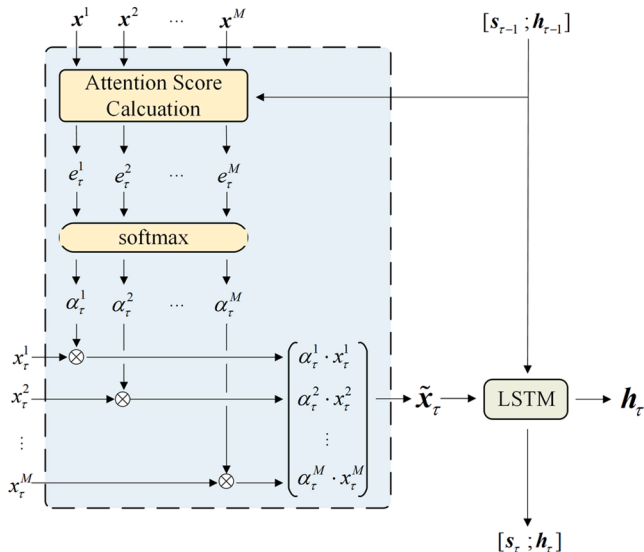


Fig. 1. The structure of feature attention based encoder.

from Eq.1 to Eq.5, and x_t is replaced by the weighted features representation $\tilde{x}_t$.

2.2. Temporal attention based decoder

Just like the number of features, the time scale of historical data is also a factor affecting the performance of times series regression. To find out the key time points of the historical data, we incorporate temporal dependencies by both the LSTM network and temporal attention mechanism in the decoder. The LSTM network stores the historical information and keeps long-term information, while the temporal attention selects parts of hidden states across all time steps.

The temporal attention mechanism is implemented by a two-layer perceptron, followed by an LSTM network. The structure of the temporal attention based decoder is shown in Fig. 2. For each encoder output h_t , its temporal attention score l_t^r is computed as

$$l_t^r = V_d^T \tanh(W_d[d_{t-1}; s_{t-1}^d] + U_d h_t + b_d) \quad (10)$$

referring to the hidden layer state d_{t-1} and the cell state s_{t-1}^d at the previous time step of the decoder LSTM network, where $V_d \in \mathbb{R}^r$, $W_d \in \mathbb{R}^{r \times 2r}$, $U_d \in \mathbb{R}^{r \times p}$ and $b_d \in \mathbb{R}^r$ are the weights and bias of the two-layer perceptron in the decoder, and r is the number of hidden units in the last layer of the decoder LSTM network.

The temporal attention weight

$$\beta_t^r = \text{softmax}(l_t^r) = \frac{\exp(l_t^r)}{\sum_{j=1}^T \exp(l_j^r)} \quad (11)$$

is also the softmax normalization form of the score l_t^r , which stands for the importance of the τ -th encoder output at the time point t .

To synthesize the temporal information across all time steps, the context vector is the weighted summation of all encoder outputs the following:

$$c_t = \sum_{\tau=1}^T \beta_t^r h_\tau \quad (12)$$

Thus, each context vector at the time step t , $c_t \in \mathbb{R}^{p \times 1}$ involves both the feature and the temporal attention results on the input feature matrix.

It should be noted that the context vector contains merely the information of the input features. Therefore, we combine the context vector c_t with y_t by matrix concatenation to incorporate the target series in the following expression:

$$\tilde{y}_t = \tilde{w} [y_t; c_t] + \tilde{b} \quad (13)$$

where $\tilde{w} \in \mathbb{R}^{1 \times (p+1)}$ and $\tilde{b} \in \mathbb{R}$ is the weight and bias of the fully con-

nected layer in the decoder. The concatenation results are then passed through a single fully connected layer as the input to the decoder LSTM network.

After T times iteration, the time information is collected comprehensively through the LSTM layer, with key time points enhanced by the temporal attention mechanism. We denote the final hidden state d_T of the decoder LSTM network as the decoder output and obtain

$$d_t = f_{LSTM}^d(d_{t-1}, \tilde{y}_t) \in \mathbb{R}^{1 \times r} \quad (14)$$

$$f_t^d = \sigma \left(W_f^d [d_{t-1}; \tilde{y}_t] + b_f^d \right) \quad (15)$$

$$i_t^d = \sigma \left(W_i^d [d_{t-1}; \tilde{y}_t] + b_i^d \right) \quad (16)$$

$$o_t^d = \sigma \left(W_o^d [d_{t-1}; \tilde{y}_t] + b_o^d \right) \quad (17)$$

$$s_t^d = f_t^d \odot s_{t-1}^d + i_t^d \odot \tanh \left(W_s^d [d_{t-1}; \tilde{y}_t] + b_s^d \right) \quad (18)$$

$$d_t = o_t^d \odot \tanh(s_t^d) \quad (19)$$

where f_{LSTM}^d represents the update function of the LSTM network in the decoder. In detail, W_f^d , W_i^d , W_o^d and b_f^d , b_i^d , b_o^d are the gate weight matrices and the biases of the forget, the input and output gates, respectively; W_s^d and b_s^d are the weight matrix and the bias of the memory cell. The sigmoid activation function σ is chosen for the gate activation.

The final predicted value $\hat{y}_{T+1}$ is determined by the decoder output d_T and the context vector c_T . These two vectors are concatenated firstly and then forwarded through two fully connected layers for the dimension transformation as the following:

$$\hat{y}_{T+1} = V_y^T (W_y [d_T; c_T] + b_w) + b_y \quad (20)$$

where $W_y \in \mathbb{R}^{r \times (p+r)}$, $V_y \in \mathbb{R}^r$ and $b_w \in \mathbb{R}^r$, $b_y \in \mathbb{R}$ are the weights and biases of the two fully connected layers.

3. Short-term probabilistic load forecasting framework

The forecasting framework of the dual-stage attention based LSTM model is presented in Fig. 3. Denote the target series as the historical records of the electricity load $Y = (y_1, y_2, \dots, y_T)$ and the nearby weather stations information as the feature matrix $X = (x_1, x_2, \dots, x_T) = (x_1^1, x_2^1, \dots, x_T^M)^T$; the prediction model F_Ω is to be learned to forecast the value of electricity load $\hat{y}_{T+1}$ at the next time point in Eq. (21).

$$\hat{y}_{T+1} = F_\Omega(y_1, y_2, \dots, y_T, x_1, x_2, \dots, x_T) \quad (21)$$

where T is the number of the time steps for iterations.

In the encoder, the feature attention mechanism calculates the attention weight α_t^m to show the significance of the m -th weather station on electricity load at time step τ . Then weighted feature information $\tilde{x}_t$ is fed through the encoder LSTM network.

Then the decoder is designed to focus on certain encoder output h_t through the temporal attention weight β_t^r at each time step. The context vector c_t and the previous load value are concatenated together and put through the decoder LSTM network. Then through the dimension transformation, the predicted load value $\hat{y}_{T+1}$ is obtained.

The traditional model using root mean square error (RMSE) as loss function can merely output single value for decision-making. In order to capture the uncertainty of load fluctuations in the future, we replace the RMSE by the pinball loss between the predicted quantile value $\hat{y}_{T+1}$ and real value y_{T+1} as the loss function.

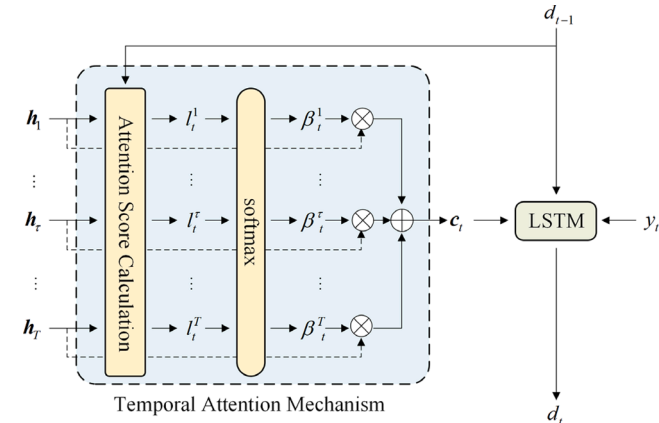


Fig. 2. The structure of temporal attention based decoder.

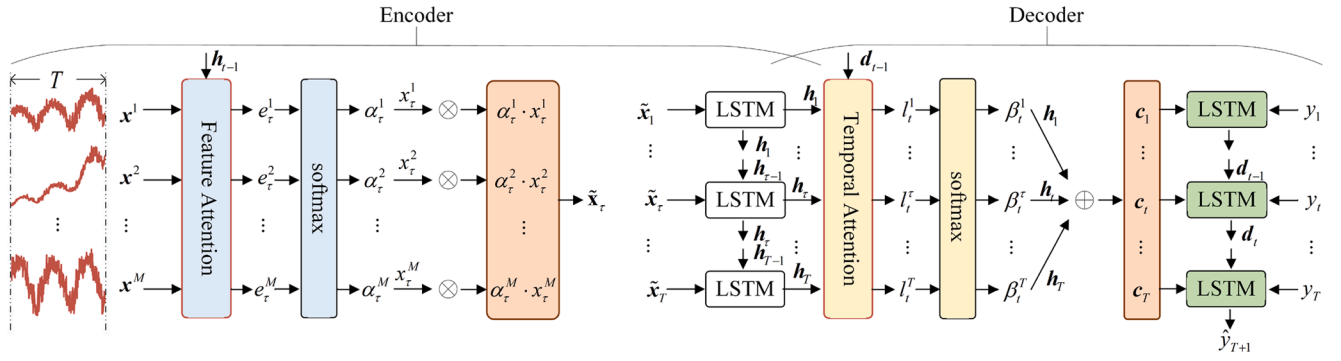


Fig. 3. Framework of the dual-stage attention based LSTM.

$$L(\hat{y}_{T+1}, y_{T+1}, \Omega, q) = \sum_{i: \hat{y}_{T+1} < y_{T+1}^i} (1 - \frac{q}{100})(y_{T+1}^i - \hat{y}_{T+1}) + \sum_{i: \hat{y}_{T+1} \geq y_{T+1}^i} \frac{q}{100}(y_{T+1}^i - \hat{y}_{T+1}) \quad (22)$$

where Ω is the set of all parameters in our model to learn; i is the sample index training dataset; q is the predefined quantile value, ranging from 0 to 100.

As the quantile value q varies, a series of predicted values at the same time point could be plotted to reveal the uncertainties. From the scattered points, it is vivid for us to evaluate the probabilistic results on reliability, sharpness and resolution [1].

Therefore, the LSTM works as the core encoder-decoder sequence structure for time step iteration. Two attention mechanisms are embedded respectively into the encoder and the decoder to select relevant weather stations and key historical time points to the electricity load. The whole network is trained by the quantile loss to estimate the uncertainties of the prediction under the different quantiles.

4. Evaluation indexes and benchmarks

4.1. Experimental setup

Several state-of-the-art load forecasting methods which have been widely taken as benchmarks in the load forecasting studies are selected here to compare their performances against the proposed approach. Herein, the quantile random forest (QRF) is noted as M1, the quantile multilayer neural network (QMNN) as M2, the quantile gradient boosting regression tree (QGBRT) as M3, the quantile long short-term memory network (QLSTM) as M4, the feature attention based quantile long short-term memory network (FA-QLSTM) as M5, the temporal attention based quantile long short-term memory network (TA-QLSTM) as M6. The proposed DA-QLSTM as M7. More specifically, the hyperparameters of the proposed DA-QLSTM are listed in Table 1. For a fair comparison, the number of LSTM layers in QLSTM is set as 6, and each layer has 32 hidden units. The FA-QLSTM and TA-QLSTM are set by only

Table 1
Hyperparameters of the proposed DA-QLSTM.

Parameter description	Value
Feature attention based encoder	3 LSTM layers; 32 hidden units in each layer
Temporal attention based decoder	3 LSTM layers; 32 hidden units in each layer
Mini-batch size	256
Optimizer	Adam
Learning rate	0.001
Dropout probability	0.1
Number of epochs	200

using one type of attention mechanism of the proposed model while keeping the other hyperparameters the same. The number of hidden layers in QMNN is set as 6, and each layer has 256 hidden units.

All the tested algorithms were run on a DELL PC with an NVIDIA GeForce GTX 1650 4 GB GPU and a 2.6 GHz Intel i7 processor. These were implemented in Python using the packages of scikit-learn [29], Keras [30] and Tensorflow [31].

4.2. Performance evaluation

4.2.1. Point forecasting performance

The mean absolute error (MAE), the mean absolute percentage error (MAPE), RMSE and the normalized root mean square error (NRMSE) are frequently used error measures in point forecasting, which can be expressed as follows:

$$MAE = \frac{1}{N} \sum_{t=1}^N |L_t - \hat{L}_{t,q=50}| \quad (23)$$

$$MAPE = \frac{1}{N} \sum_{t=1}^N \left| \frac{L_t - \hat{L}_{t,q=50}}{L_t} \right| \times 100\% \quad (24)$$

$$RMSE = \sqrt{\frac{\sum_{t=1}^N (L_t - \hat{L}_{t,q=50})^2}{N}} \quad (25)$$

$$NRMSE = \frac{\sqrt{\frac{1}{N} \sum_{t=1}^N (L_t - \hat{L}_{t,q=50})^2}}{L_{max} - L_{min}} \times 100\% \quad (26)$$

where L_t and $\hat{L}_{t,q=50}$ are the actual load data and the 50th percentile quantile value of the predicted load at the time step t , respectively. The $\hat{L}_{t,q=50}$ can be obtained by setting the $q = 50$ in when training the proposed model. N is the total number of the test data points. L_{max} and L_{min} are the maximum and the minimum value of the actual load data.

4.2.2. Probabilistic forecasting performance

The probabilistic forecasting performance can be assessed from different aspects including consistency, variation and the tightness of the estimated distribution. The pinball score is one of the most comprehensive and widely used metrics that can evaluate the calibration, reliability, and sharpness of the model [1]. It can be calculated as the mean of pinball losses throughout the whole forecast horizon and across all set quantiles, which is shown as follows:

$$L_{t,q}(L_t, \hat{L}_{t,q}, q) = \begin{cases} (1-q)(\hat{L}_{t,q} - L_t) & \hat{L}_{t,q} \geq L_t \\ q(L_t - \hat{L}_{t,q}) & \hat{L}_{t,q} < L_t \end{cases} \quad (27)$$

$$Pinball = \frac{1}{N \times |Q|} \sum_{t=1}^N \sum_{q \in Q} L_{t,q}(L_t, \hat{L}_{t,q}, q) \quad (28)$$

where $\hat{L}_{t,q}$ is the q^{th} quantile of the predicted value at time t and $Q = \{1, 2, \dots, 99\}$ is the quantile set. $|Q|$ denotes the cardinality of Q . A lower pinball score value indicated better performance.

The Winkler score (WS) score is another type of measurement metric for the probabilistic forecasting that evaluates the performance within interval width. The WS for a central $(1 - \alpha) \times 100\%$ prediction interval (PI) is defined as follows:

$$WS = \begin{cases} \Delta_t + 2(L_t - \max_t)/\alpha & L_t > \max_t \\ \Delta_t & \min_t < L_t < \max_t \\ \Delta_t + 2(\min_t - L_t)/\alpha & L_t < \min_t \end{cases} \quad (29)$$

where $\max_t$ and $\min_t$ represent the upper and lower bounds of the probabilistic forecasts at time t , respectively; $\Delta_t = \max_t - \min_t$ is the width of PI. A lower WS score infers a superior probabilistic performance regarding the certain interval.

5. Case studies

5.1. Dataset description

The dataset of GEFCom2014 [32] is chosen for the case study in this paper. This dataset includes seven years of hourly electricity load data from 2005 to 2011 and ten years of hourly temperature data from 2001 to 2011 measured by 25 weather stations. The load data from 2006 to 2010 and the temperature data are used for training the forecasting model. These data are split into a training set and a validation set with the portion of 90% and 10%, respectively. Year 2011 is used for out-of-sample test.

5.2. Point and probabilistic forecasting results

This section analyzes the forecasting performance of the proposed DA-QLSTM method and compares the result with other methods in terms of both the point and the probabilistic forecasting. Fig. 4 shows the one hour ahead probabilistic forecast results of 7 test days (1 week) from June 20, 2011 to June 26, 2011 obtained via the proposed DA-QLSTM model. The actual load during the test period is presented in the black curve with cross and the red line denotes the 50th quantile predicted values. The PI ranges from 10% to 90% with 20% incremental change each are indicated by a decreasing color depth of the blues. Visually inspecting the figure, the quantiles can effectively capture the basic trends in the load profile with a tight prediction coverage interval. In the entire test period, the proposed method can achieve superior performance for zonal load forecasting, and there is no direct correlation between the forecast error and time that can be observed. This is because the extracted hidden representations can well fit the future dynamics by referring to the most relevant temporal pattern in the historical data.

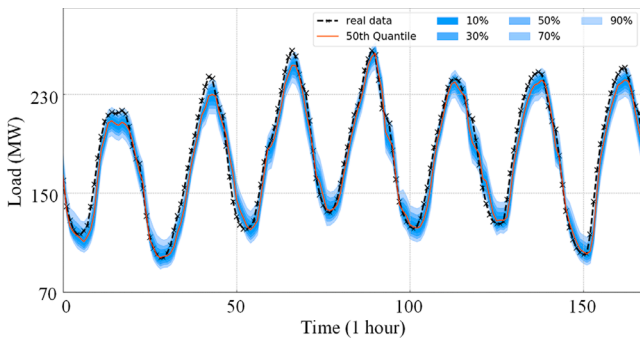


Fig. 4. Probabilistic forecasts results of DA-QLSTM model.

Suitable feature extractions can always improve the model's forecasting ability. It is significant to underline that the load during the peak hours of each day can be well predicted within a reliable range utilizing the DA-QLSTM model. This particular capability is of vital importance for power system planning and operation.

Fig. 5 illustrates the point forecasting result of different methods represented with bars. A higher value corresponds to a longer bar. It shows that the proposed DA-QLSTM model M7 has the most prominent point forecasting performance, as indicated by the approximately 50.3%, 47.0%, 49.8% and 50.4% lower RMSE, MAE, MAPE and NRMSE, respectively, when compared with neural network regression (M2). Furthermore, when compared with the other two LSTM network considering only one attention mechanism (M5 and M6), the performance of DA-QLSTM model also stands out with at least 15.6%, 10.5%, 26.4% and 15.7% improvements in the four evaluation metrics. This is because the proposed DA-QLSTM can mine the correlations between the features spatially and extract the dependencies within a certain period of time temporally, thereby improving the forecasting performance.

The pinball scores and Winkler scores given in Fig. 6 are used to evaluate the probabilistic forecasting performance of the different models. They illustrate that the DA-QLSTM model still outperforms all the other methods significantly with approximately 44.4%, 46.4% and 35.1% performance enhancements regarding all the evaluation metrics, respectively, compared with M1. Also shown in Fig. 6, M2 has poor performance probably because it has overfitting issues in a random forecast, which is sensitive to noise data. The performance of DA-QLSTM also dominates when compared with the best of the state-of-art methods (M6), with approximately 14.6%, 11.9% and 10.2% improvements, respectively, in the three probabilistic evaluation metrics. Since a main limitation of the machine learning-based method is that it has limited degree of explicability. To evaluate the quality of extracted features, one direct and simple way is to compare the forecasting performance of different models from multiple perspectives. Generally speaking, when the forecasting outputs of a model have high accuracy, it means the corresponding input features have strong characterization ability. Comprehensively comparing the evaluation metrics shown in Fig. 5 and Fig. 6, it can be seen that the DA-QLSTM model stands out both in the point and the probabilistic forecasting, in terms of reliability, sharpness and resolution. These observations can fully demonstrate the effectiveness of the proposed method to use the attention based encoder-decoder architecture to conduct feature extraction.

One significant factor affecting the accuracy of the model is the number of hidden units in the network. The tuning of the number of hidden units is difficult in the stage of hyperparameter tuning if the network is sensitive to the number of hidden units. In order to investigate how and to what extent the number of hidden units can influence the final load forecasting results, the proposed DA-QLSTM model is tested by setting different numbers of hidden units. The Pinball score and training time are shown in Fig. 7. As shown, when increasing the number of hidden units, the Pinball score firstly decreases quickly and then becomes stable, while the training time continuously increases. Although increasing the number of hidden units can improve the model performance, the model training would take more time. Jointly

Methods \ Metrics	RMSE(MW)	MAE(MW)	MAPE(%)	NRMSE(%)
M1 (QRF)	10.57	8.31	5.74	6.28
M2 (QRNN)	9.48	7.38	4.50	5.65
M3 (QGBRT)	8.90	6.42	4.09	5.32
M4 (QLSTM)	6.17	5.29	3.85	3.69
M5 (FA-QLSTM)	5.95	4.42	3.18	3.55
M6 (TA-QLSTM)	5.58	4.37	3.07	3.32
M7 (DA-QLSTM)	4.71	3.91	2.26	2.80

Fig. 5. Point forecasting performance of different models.

Metrics Methods	Pinball	WS(50%)	WS(90%)
M1 (QRF)	4.05	30.61	52.48
M2 (QMNN)	2.22	22.82	34.49
M3 (QGBRT)	2.91	25.37	40.58
M4 (QLSTM)	1.98	18.72	30.11
M5 (FA-QLSTM)	1.56	15.91	26.30
M6 (TA-QLSTM)	1.44	13.87	24.92
M7 (DA-QLSTM)	1.23	12.22	22.39

Fig. 6. Probabilistic forecasting performance of different models.

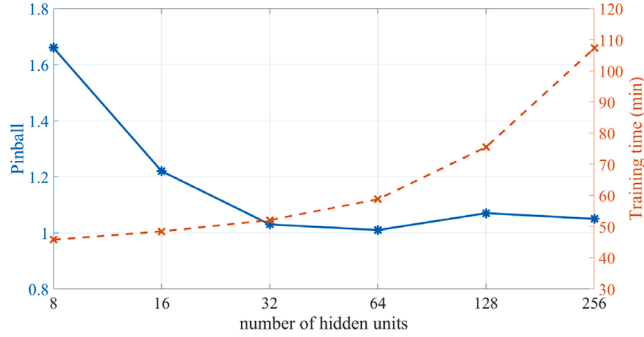


Fig. 7. Pinball score and training time for proposed model with different number of hidden units.

considering the forecasting accuracy and model complexity, we set the number of hidden units as 32 in this paper.

Moreover, we examine the impact of forecast time interval on the proposed DA-QLSTM model. Table 2 presents three probabilistic evaluation results of varying time scale, defined by T_{test} . As observed from Table 2, the forecast errors will increase if the forecast horizon becomes longer. For example, the model predicting the next 6 h load data ($T_{test} = 6$ h) is extensively superior to the case of $T_{test} = 2$ weeks, with relative improvements of 45.2%, 36.2% and 39.6% for the Pinball scores, WS (50%) and WS(90%), respectively. According to the scope of the evaluation results from Table 2, we can further divide the prediction time scale into three tiers, namely less than 1 day, within 1 week and within 2 weeks. A trade-off between the prediction accuracy and the length of forecast time scales can be made based on the system operator's requirements.

Another case study lies in demonstrating the superior stability and generalization performance of the proposed method. We calculate the MAPE, Pinball scores and Winkler scores for each day in a month. Then we draw the box-plots for the distribution of each metric for the hourly period in a day. The distributions are shown in Fig. 8. The pinball score and Winkler score have similar trends under different moment in a day, while the distribution for MAPE is different. The outliers of three evaluation metrics are more spread from 6:00 to 8:00 and from 15:00 to 22:00. These two time periods correspond to people go to and after work. There exist large uncertainties because of higher load demands consumed by end users, which lead the load more difficult to predict.

Table 2
Results of different forecasting time interval.

Time interval	Pinball	WS(50%)	WS(90%)
6 h	0.89	10.2	15.51
12 h	1.1	10.01	16.62
1 day	1.09	11.16	19.42
2 days	1.17	12.21	18.92
3 days	1.25	12.69	20.19
1 week	1.22	12.22	22.39
2 weeks	1.46	15.98	25.71

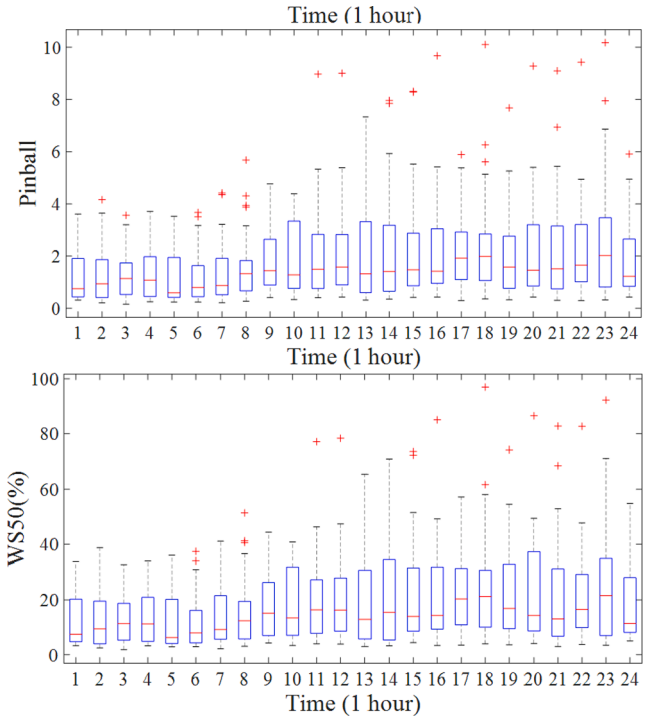


Fig. 8. Distribution of evaluation metrics at different time in a day.

Table 3
Training time of different methods.

Methods	Training time (minute)
M1 (QRF)	19.32
M2 (QRNN)	38.74
M3 (QGBRT)	11.41
M4 (QLSTM)	34.28
M5 (FA-QLSTM)	43.77
M6 (TA-QLSTM)	46.58
M7 (DA-QLSTM)	52.09

In respect of the computational complexity, the training time of all the examined models are listed in Table 3. Although it takes longer time for the proposed DA-QLSTM to complete the model training than the other comparison methods, the entire training process is executed off-line. Given the trained DA-QLSTM, it is noted that only a few seconds are needed to output the forecasting results in practical applications. Therefore, the aim in our paper is to build a model to obtain the accurate forecasting results.

5.3. Robustness analysis

In practical applications, bad data and missing data are inevitable when collecting the measurements of zonal load. In this part we aim to explore the proposed DA-QLSTM model's robustness to the prediction accuracy of these two issues.

1) Outliers: First, we investigate how the number of outliers affects the accuracy of the proposed model. The outliers are those points that significantly deviate from the normal observations. They are mainly caused by the communication errors or disruptions in the measurements. In this case, the outliers are generated by adding Gaussian noises $N(0, \sigma^2)$ with a large value of standard deviation σ to the original data. Here, we set σ to a quarter of the original value. We randomly modified different rates of training samples as outliers, and the probabilistic forecasting performance for all the examined methods trained on these modified training datasets are shown in Fig. 9. It can be seen that when

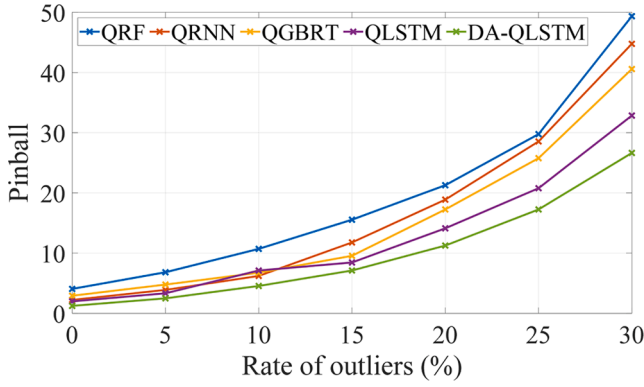


Fig. 9. Pinball scores with varying rate of outliers.

there are a small number of outliers for model training, the proposed DA-QLSTM can still maintain a higher forecasting accuracy than the benchmark methods. However, when invading a huge number of anomalies, the model performance deteriorates significantly. For example, when the rate of outliers is 30%, the Pinball score reaches nearly 10 times the value that is free of any irregular measurements, meaning that all the models fail to prevent credible predictions under this condition. In general, the proposed method has satisfactory performance and strong robustness within a certain degree of outliers.

2) Data incompleteness: The existing missing data is another commonly encountered issue for the time series prediction problem. In this case, we simulate the data deficiency by randomly discarding different rates of training samples, and then fill up the missing positions with zeros. Fig. 10 presents the pinball scores for all the methods. It can be seen that when the minority of data (less than 10%) is missing for all models, their forecasting performance is slightly affected. As the number of missing data grows, their forecasting performance is increasingly impacted, especially for the QRF model. Overall, our proposed method still outperforms the other comparison methods under all conditions, which demonstrates its superior robustness to the data incompleteness issue.

5.4. Attention results and weather station selection

It has been demonstrated by empirical practices that the weather is a major factor affecting the electricity demand. The weather variables such as the temperature, the humidity, the cloud cover and the wind speed, are often considered as the input features of the model. As reported in [27], an area usually contains multiple weather stations geographically. It is not practical to manually investigate how to assign the best weather stations. A novel model is now proposed to decide how many and which weather station to use for a specific territory. The proposed DA-QLSTM model can effectively solve this problem. It can automatically learn the importance of each variable at each time step in

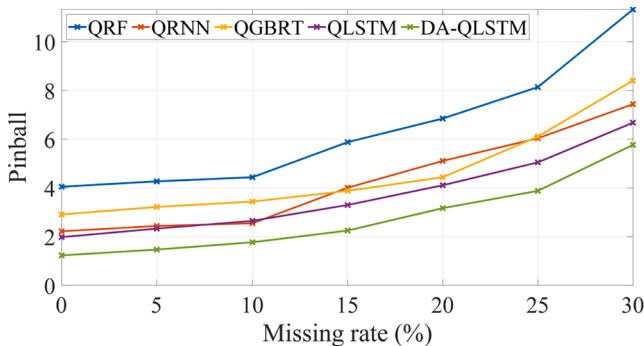


Fig. 10. Pinball scores with varying rate of missing data.

the training process, and make the most critical components play a leading role by weighing each input feature.

In order to better show the internal states of attention mechanism, the heatmaps of dual-stage attention weights ($\times 10^{-3}$) is presented in Fig. 11. The higher degree of influence corresponds to the darker color. It visually shows the trend of influence degree of temperature data from different weather stations on the forecast point as the time step changes. The most influential weather station is #18, with weight approximately 80% higher than the average and nearly three times that compared with the weather station #25. In addition, the temperature features collected from each weather station have different weights at each time step on the prediction point at time t . The most recent look-back point has the most influence on the prediction point. The proposed DA-QLSTM model can quantify the impact of each input feature on the object forecast dynamically varying over time. With the attention mechanism, the correlation results can be calculated, which is instructive for feature dimension reduction.

Based on the attention weights, we select the temperature data from the top 4 most influential weather stations (#12, #14, #18 and #20) as the inputs to the model. The point and probabilistic forecasting results of DA-QLSTM before and after the feature reduction are given in Table 4. As can be seen, with the same network architecture, the forecasting accuracy of DA-QLSTM with 4 higher-correlated weather stations is slightly improved in terms of the evaluation metric values. It should be pointed out that the trade-off between the forecasting performance and feature selection should be decided by the system operators. For example, if the operators can accept the less than 5% lower pinball loss value, the relevant feature selection process which is necessary for the traditional methods, can be avoided for our proposed method, thus relieving them from intensive work on the feature engineering. Even without complex feature engineering, the proposed model can still obtain accurate forecasting results, since the proposed attention mechanism can automatically enhance the impact of higher correlation features and reduce the interference of the unrelated variables.

5.5. Different temperature scenarios analysis

Prior studies have already shown that the ambient temperature could have considerable influences on the electricity load. Many different temperature scenario generation methods for model training have been reported in the load forecasting literature, which can be roughly divided into following categories: the fixed-date method, the shift-date method, the bootstrap method and the surrogate method [14]. As proposed in [14], a rule of thumb can be given to select appropriate parameters for effective temperature scenario generation:

$$n = \lceil 30/k - 1 \rceil \quad (30)$$

where k denotes the number of years of historical temperature data, and n is the number of days the temperature profile being shifted around. For examples, if we have 4 years of temperature history, we should shift about 7 days forward and backward to generate more temperature scenarios during training process.

Based on this finding, we intend to investigate whether the performance of proposed DA-QLSTM model would be affected by different generated temperature scenarios. We set the basic model with 4 important weather stations data as Scenario 0, and consider three other generated temperature scenarios.

Scenario1: Shift $k = 2$ years and $n = 14$ days around

Scenario2: Shift $k = 3$ years and $n = 9$ days around.

Scenario2: Shift $k = 4$ years and $n = 7$ days around.

Fig. 12 presents the pinball scores and the bar plots for all forecasting approaches. As shown, although the performance of M1-M6 have been improved when the extra years of historical temperature data is considered to train the model, the proposed DA-QLSTM (M7) consistently outperforms the other approaches. The QGBRT model (M3) has

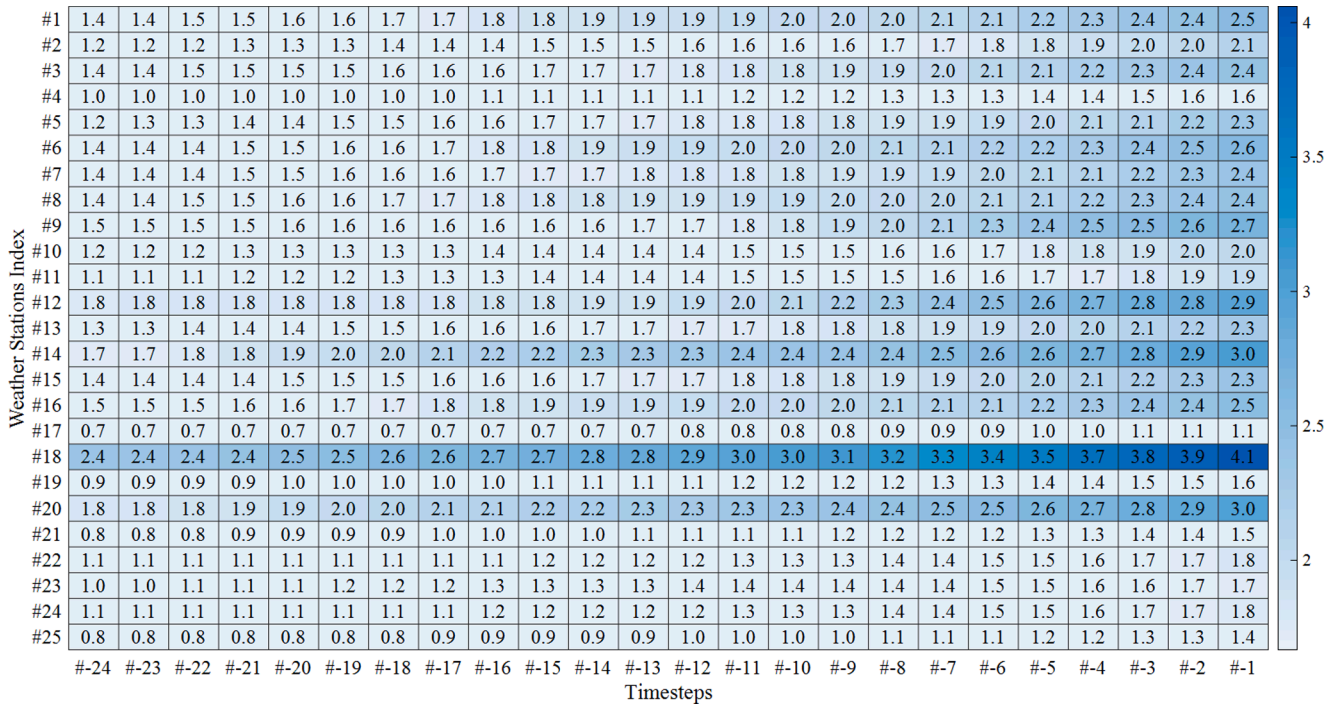


Fig. 11. Heatmap of dual-state attention weights for the 25 weather stations.

Table 4

Point and probabilistic forecasting results of DA-QLSTM before and after weather station selection.

	25 weather stations	4 weather stations	Relative improvement (%)
RMSE (MW)	4.71	4.69	0.42
MAE(MW)	3.91	3.73	4.6
MAPE(%)	2.26	2.13	5.75
NRMSE(%)	2.8	2.79	0.36
Pinball	1.23	1.19	3.25
WS(50%)	12.22	12.25	-0.25
WS(90%)	22.39	20.16	9.96

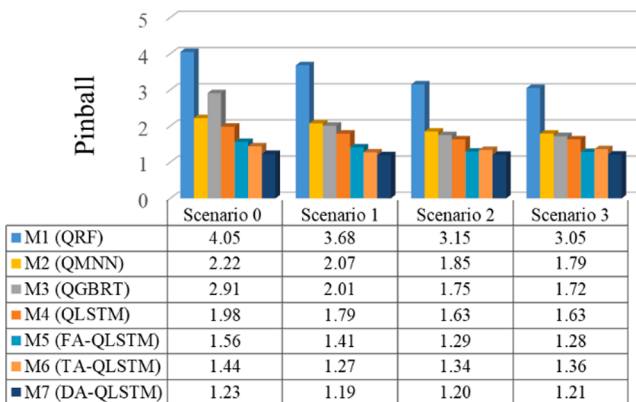


Fig. 12. Pinball scores under different generated temperature scenarios.

the biggest improvement with a relative increase of 40.9% in Scenario 3 when compared with Scenario 0. Apart from this, the relative improvement of M7 is comparatively small, showing approximately 3.3%, 2.4% and 1.6% lower pinball score, respectively, compared with Scenario 0, indicating that the proposed model can obtain superior prediction effect even without adding more generated temperature

scenarios.

Additionally, we further analyze how the model would perform when using the forecast temperature data as inputs during the test stage. Since the temperature data utilized for the input are all real data, the forecast results obtained previously are the upper bound of the performance of the model [17]. We modified the original temperature values in the test stage by adding Gaussian noise in order to examine whether the model is robust to noise in forecast weather.

Scenario 1: mean 0 °F, and standard deviation 1 °F.

Scenario 2: mean 0 °F, and standard deviation 2 °F.

Scenario 3: mean 0 °F, and standard deviation 3 °F.

The pinball scores of each model with modified temperature values are shown in Fig. 13. As can be seen in the figure, with the increasing noise adding to the temperature data, the probabilistic load forecasting results of all the tested methods decrease, indicated by the increased pinball scores, especially for Scenario 3, where the maximum growth rate is 119.2% of M3. The proposed DA-QLSTM outperforms all the other methods in every scenario with the smallest increase of pinball scores. It is reasonable to conclude that the proposed model is robust against the uncertainty of temperature and has higher generalization capability. There are two principal reasons for this robustness. Firstly, we use a feature attention based encoder to extract the driving input

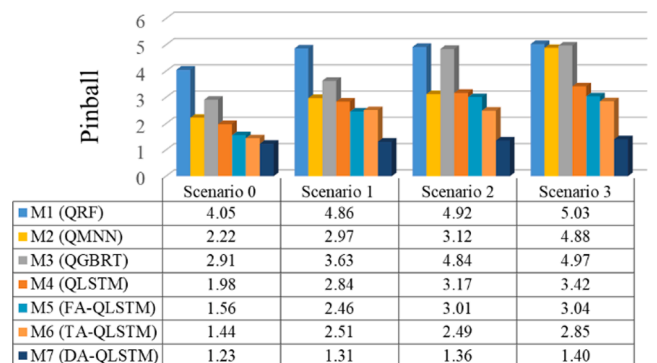


Fig. 13. Pinball scores under different modified temperature scenarios.

series which can better reveal the correlation between the raw input features and the output zonal load. The second is that the different extracted dynamic features are then modeled discriminately by using a temporal attention based decoder to select relevant hidden features across all time steps. The integration of these two attention mechanisms empowers the proposed method to learn proper latent representations and thus accurate forecasts can be obtained.

6. Conclusion and future work

In this paper, a novel dual-stage attention based LSTM network is proposed for short-term zonal load probabilistic forecasting, which improves prior work particularly with respect to multiple sources of data and generalization. A feature attention based encoder and a temporal attention based decoder are developed to evaluate the correlation of the input exogenous features with the load data and discover the temporal dependencies, respectively. This framework can adaptively enhance the influence of higher correlation features on the forecast points, and reduce the impact of irrelevant variables. The open dataset of GEFCom2014 is used to verify the effectiveness of proposed model with numerous test cases. The proposed method outperforms in both the point and the probabilistic forecasting with the higher accuracy and the robustness to the temperature variation compared with other state-of-the-art forecasting models. We also demonstrate that the weights of attention mechanism are instructive for variable selection, and can help solve the problem of finding the most important features. The proposed model can still provide accurate and reliable forecasting results without variable selection and feature engineering, which is easy to use by engineers.

The future work will include two directions. Since the DRES, which are mostly installed behind the meters, may cause uncertainties on the zonal load forecasting, the first future work is to separate the behind-the-meter resources and extend our model to consider the impact of these uncertainties of the DERS. Since the humidity data is not included in the GEFCom2014 dataset and only the temperature data is considered in this paper, the other future work will investigate the effects of the humidity data on the zonal load forecasting accuracy using our proposed method.

CRedit authorship contribution statement

Jun Lin: Writing – review & editing, Methodology, Software, Validation. **Jin Ma:** Writing – review & editing, Conceptualization. **Jianguo Zhu:** Writing – review & editing, Conceptualization. **Yu Cui:** Visualization, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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